

One-Way ANOVA

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Introduction

One-way ANOVA tests whether the means of three or more independent groups are equal. It generalises the two-sample t-test to multiple groups while controlling overall Type I error. The omnibus F-test tells us whether *any* means differ; post-hoc tests (Tukey HSD, Dunnett) identify which pairs.

Prerequisites

Two-sample t-test, F distribution.

Theory

For k groups with n_i observations each, total $n = \sum n_i$, define:

- **Sum of squares between** (SSB): $\sum_i n_i (\bar{X}_i - \bar{X})^2$ with $k - 1$ df.
- **Sum of squares within** (SSW): $\sum_i \sum_j (X_{ij} - \bar{X}_i)^2$ with $n - k$ df.
- **Mean squares**: $MSB = SSB / (k - 1)$, $MSW = SSW / (n - k)$.

Under $H_0 : \mu_1 = \dots = \mu_k$,

$$F = \frac{MSB}{MSW} \sim F_{k-1, n-k}.$$

Large F implies between-group variation exceeds within-group variation, rejecting equality of means.

Effect size: $\eta^2 = SSB/SST$, the proportion of variance explained by the group factor.

Assumptions

- Independent observations within and across groups.
- Normal residuals (or large enough n_i).
- Homogeneous variances across groups (Welch ANOVA relaxes this).

R Implementation

```
library(effectsize); library(broom); library(car)
set.seed(2026)

df <- data.frame(
  group = factor(rep(c("A", "B", "C", "D"), each = 30)),
  y      = c(rnorm(30, 50, 8),
             rnorm(30, 55, 8),
             rnorm(30, 58, 8),
             rnorm(30, 54, 8))
)

# Omnibus ANOVA
```

```

fit <- aov(y ~ group, data = df)
summary(fit)
tidy(fit)

# Assumption checks
leveneTest(y ~ group, data = df)
shapiro.test(residuals(fit))

# Effect size
eta_squared(fit)
omega_squared(fit)

# Welch ANOVA (unequal variances)
oneway.test(y ~ group, data = df)

# Tukey HSD post-hoc
TukeyHSD(fit)

```

Output & Results

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	3	891	297.0	4.83	0.00317 **
Residuals	116	7135	61.5		

Eta2 (partial) | 95% CI

0.11 | [0.02, 1.00]

Tukey multiple comparisons of means 95% family-wise confidence level

	diff	lwr	upr	p adj
B-A	5.421	-0.58	11.42	0.097
C-A	8.215	2.21	14.22	0.003
D-A	4.129	-1.87	10.13	0.287
C-B	2.794	-3.21	8.79	0.627
D-B	-1.292	-7.29	4.71	0.943
D-C	-4.086	-10.09	1.92	0.297

$F(3, 116) = 4.83$, $p = 0.003$, rejects equal means. Tukey identifies C vs. A as the significant pair.

Interpretation

Report: “A one-way ANOVA showed a significant group effect ($F(3, 116) = 4.83$, $p = 0.003$, $\eta^2 = 0.11$). Tukey HSD comparisons indicated that group C differed significantly from group A (mean difference 8.2, 95 % CI 2.2 to 14.2, adjusted $p = 0.003$).”

Practical Tips

- Always check homogeneity of variances; if violated, use Welch ANOVA.
- Always report an effect size; F alone is not enough.
- For many post-hoc comparisons, Tukey is the standard; Dunnett is for comparing treatments to a control; Bonferroni or Holm for pre-specified families.
- Non-parametric alternative: Kruskal-Wallis.
- For within-subjects data, use repeated-measures ANOVA or a linear mixed model.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests